# Tutorial 3

## Exercise 1

There are four monetary outcomes $x \in \{0, 20, 50, 100\}$, in euros. A lottery is a vector $L = (p_0, p_{20}, p_{50}, p_{100})$ of probabilities, and $\delta_x$ is the lottery that pays $x$ for sure. A subject's preferences $\succsim$ over lotteries satisfy the conditions of the expected utility theorem, so there exist numbers $u(0), u(20), u(50), u(100)$ with

$$L \succsim L' \iff \sum_x p_x\, u(x) \geq \sum_x p'_x\, u(x).$$

We normalise $u(0) = 0$ and $u(100) = 1$. In an experiment the subject reports two indifferences:

$$\delta_{20} \sim (0.6,\ 0,\ 0,\ 0.4), \qquad \delta_{50} \sim (0.3,\ 0,\ 0,\ 0.7).$$

(a) Find $u(20)$ and $u(50)$.

(b) Rank the following lotteries for this subject, and compare your ranking with the ranking by expected value:

$$A = (0.1,\ 0.3,\ 0.3,\ 0.3), \qquad B = \delta_{50}, \qquad C = (0.35,\ 0,\ 0,\ 0.65).$$

(c) Find the certainty equivalent and the risk premium of the lottery that pays €20 or €100 with probability $\frac{1}{2}$ each.

## Exercise 2

A worker earns $w = 100$. With probability $\pi$ they are laid off, in which case their income falls by $D = 60$. They can buy unemployment insurance: coverage $a \in [0, D]$ costs $qa$, paid whether or not they are laid off, and pays $a$ if they are laid off. Their consumption is therefore

$$c_1(a) = w - qa \ \ \text{(employed)}, \qquad c_2(a) = w - D + (1 - q)a \ \ \text{(laid off)}.$$

Their Bernoulli utility $u$ is strictly increasing, strictly concave and differentiable, and they maximise expected utility $V(a) = (1 - \pi)\, u(c_1(a)) + \pi\, u(c_2(a))$. Throughout, $\pi = 0.05$ and $q = 0.06$. You may use the following fact without proof: if $V$ is concave and differentiable on $[0, D]$, then $a = D$ maximises $V$ on $[0, D]$ if and only if $V'(D) \geq 0$.

(a) Let $u(x) = \ln x$ and suppose they buy no insurance. Compute the mean of their income, its certainty equivalent and its risk premium. What is the most they would pay for full coverage ($a = D$)? Compare this with the actuarially fair price $\pi D$ and with the price they actually face, $qD$.

(b) For general $u$, show that $V$ is concave in $a$. Prove that full coverage is optimal if and only if $\pi \geq q$. What does this imply when insurance is actuarially fair ($q = \pi$)?

(c) Let $u(x) = \ln x$. Derive the optimal coverage

$$a^* = \frac{\pi w}{q} - \frac{(1-\pi)(w-D)}{1-q},$$

and evaluate it. Compute their certainty equivalent at $a^*$, at $a = 0$ and at $a = D$. Show that, when $q > \pi$, $a^*$ is decreasing in $w$. The coefficient of absolute risk aversion of $\ln x$ is $1/x$; relate this to your result.

(d) Now let $u(x) = -e^{-\alpha x}$ with $\alpha > 0$, whose coefficient of absolute risk aversion is constant at $\alpha$. Show that

$$a^* = D - \frac{1}{\alpha} \ln \left[ \frac{q(1-\pi)}{\pi(1-q)} \right]$$

(when this lies in $[0, D]$), evaluate it for $\alpha = 0.02$, and explain why $w$ does not appear.